
PH 301: Quantum mechanics and applications

Course Code	PH 301
Course Name	Quantum mechanics and applications
L-T-P-C	2.5-0.5-0-3
Approval	2nd Adhoc Senate (May 2012), in the Minor in Applied Physics curriculum

Course Contents

- **Module I**
- Review of postulates in quantum mechanics, observables and operators, theory of measurement in quantum mechanics, state of the system and expectation values, transition from quantum mechanics to classical mechanics-Ehrenfest theorem. [5 Lectures]
- **Module II**
- Application of Schrodinger equation in 1-D: rectangular barrier, tunneling, square potential well, delta-function potential [5 Lectures]
- **Module III**
- Basic mathematical formalism of quantum mechanics, Dirac notation, linear vector operators, matrix representation of states and operators, commutator relations in quantum mechanics, commutator and uncertainty relations, complete set of commuting observables [6 Lectures]
- **Module IV**
- Quantum computation and information: Qubits and logic gates [14 Lectures]
- **Module V**
- Theory of angular momentum in quantum mechanics, commutator relations in angular momentum, eigenvalues and eigen states of angular momentum [5 Lectures]
- **Module VI**
- Application of Schrodinger equation in 3-D models, central potentials, Schrodinger equation in spherical co-ordinates, solution to hydrogen atom problem [5 Lectures]
- **Module VII**
- Time independent non-degenerate and degenerate perturbation theory, fine-structure of hydrogen, Zeeman effect and hyperfine splitting. [5 Lectures]
- **Module VIII**
- Time dependent perturbation theory, two level systems, emission and absorption of radiation [5 Lectures]

References

1. D J Griffith, **Introduction to quantum mechanics**
2. R Liboff, **Introductory Quantum Mechanics**
3. R Eisberg & R Resnick, **Quantum physics of atoms and molecules**
4. D A B Miller, **Quantum Mechanics for Scientists and Engineers**
5. Levi, **Applied quantum mechanics**
6. B. H. Bransden & C. J. Joachain, **Quantum Mechanics**
7. J J Saku'rai, **Modern Quantum Mechanics**
8. R Shankar, **Principles of Quantum Mechanics**
9. Cohen-Tannoudji, B Diu, F Laloe, **Quantum Mechanics -Vol.1**